```vhdl
library ieee;
use ieee.std_logic_1164.all;
entity alu is
   port (reg1, reg0:                        in std_logic_vector(3 downto 0);
         enA, enSL, enSR, enM, enN: in std_logic;
         aluOut:                    out std_logic_vector(3 downto 0));
end;

architecture synth of alu is
signal a, b, c, d, e, carry:     std_logic_vector(3 downto 0);
begin
   carry(3) <= ((reg1(3) and reg0(3)) or (reg1(3) and carry(2)) or
               (reg0(3) and carry(2))) and enA;

   a(3) <=     (reg1(3) xor reg0(3) xor carry(2)) and enA;

   carry(2) <= ((reg1(2) and reg0(2)) or (reg1(2) and carry(1)) or
               (reg0(2) and carry(1))) and enA;

   a(2) <=     (reg1(2) xor reg0(2) xor carry(1)) and enA;

   carry(1) <= ((reg1(1) and reg0(1)) or (reg1(1) and carry(0)) or
               (reg0(1) and carry(0))) and enA;

   a(1) <=     (reg1(1) xor reg0(1) xor carry(0)) and enA;

   carry(0) <= (reg1(0) and reg0(0)) and enA;

   a(0) <=     (reg1(0) xor reg0(0)) and enA;

   b(3) <= reg1(2) and enSL;
   b(2) <= reg1(1) and enSL;
   b(1) <= reg1(0) and enSL;
   b(0) <= '0';

   c(3) <= '0';
   c(2) <= reg1(3) and enSR;
   c(1) <= reg1(2) and enSR;
   c(0) <= reg1(1) and enSR;

   d(3) <= reg1(3) and enM;
   d(2) <= reg1(2) and enM;
   d(1) <= reg1(1) and enM;
   d(0) <= reg1(0) and enM;

   e(3) <= (not reg1(3)) and enN;
   e(2) <= (not reg1(2)) and enN;
   e(1) <= (not reg1(1)) and enN;
   e(0) <= (not reg1(0)) and enN;

   aluOut <= a or b or c or d or e;
end;
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